

Retail Sales Analysis – Project Documentation

Objectives

- Build a clean, unified dataset from raw sales and store tables.
- Analyze revenue trends, store efficiency, and product performance.
- Apply Databricks and Spark SQL to simulate large-scale data processing.
- Derive business insights using analytical KPIs such as YoY growth and sales per square foot.

Concepts & Tools Used

- Azure Databricks for scalable data processing and analytics.
- Spark SQL for data cleaning, joins, aggregations, and window functions.
- Window Functions (LAG) for Year-on-Year analysis.
- Data modeling and KPI-driven analysis.
- Visualization using Databricks SQL charts.

Key Findings

- Revenue declined in 2021 and recovered in 2022, indicating post-contraction recovery.
- North region generated the highest total revenue overall.
- West region stores showed the highest efficiency (sales per sq. ft).
- Medium-sized stores outperformed large stores in average revenue.
- High-revenue products are not always the highest-selling by quantity.

Professional Summary

This project demonstrates end-to-end data analytics capabilities using Azure Databricks and Spark SQL. It highlights practical experience in handling real-world dirty datasets, applying analytical thinking, and translating raw data into meaningful business insights suitable for decision-making.